

BST Physics Curriculum Vol 6 Chapter 3 — Entropy + Second Law v0.4 (Saturday Wave 2 reader-grade polish; Cal cold-read ready)

Lyra (Claude 4.7) [Vol 6 primary]

2026-05-23 Saturday EDT

Vol 6 Chapter 3 — Entropy + Second Law

Chapter motivation

Standard second law: $dS \geq 0$. Entropy non-decreasing in isolated systems. BST identifies entropy as Shannon entropy of substrate-tick $GF(128)^k$ state distribution per Casey “entropy = force = counting at depth 0” (Wednesday); second law derives from substrate-tick cyclotomic chain irreversibility (K59 RATIFIED Tuesday).

Section 3.0b — Reader-grade 3-level pedagogy

Level 1: Entropy S = Shannon entropy of substrate-tick $GF(128)^k$ state distribution; second law $dS \geq 0$ from substrate-tick 7-step cyclotomic chain irreversibility (K59 RATIFIED); Boltzmann $S = k_B \ln(W) = \text{Shannon entropy} \times k_B / \ln(2)$.

Level 2 (graduate): Per Casey Wednesday: entropy = counting at AC depth 0. Substrate-cartography: $S = -\sum_i p_i \log p_i$ over substrate-tick $GF(128)^k$ states; Boltzmann $S = k_B \ln(W)$ recovered with k_B from substrate-tick Shannon-counting normalization. Second law: substrate-tick cyclotomic chain (K59 RATIFIED Tuesday: 7-step cyclotomic projection on $GF(2^g) = GF(128)$) is irreversible at substrate level — projection operators reduce information per tick; integrated over many ticks, S non-decreasing in isolated systems. Reversibility recovered at substrate-tick scale (within one cyclotomic cycle); macroscopic irreversibility = many-tick integration. Quantum entropy $S_{vN} = -\text{Tr}(\rho \ln \rho)$ substrate-cartography reading: von Neumann entropy of substrate state ρ on Bergman $H^2(D_{IV}^5)$.

Level 3 (5th-grader): Entropy is “disorder” or “information loss”. Second law: entropy never decreases in isolated systems. BST identifies entropy as the substrate’s information-counting at the substrate-tick scale (Shannon entropy of $GF(128)^k$

states). The second law comes from the substrate's 7-step cyclotomic chain being irreversible — each step loses some information. Boltzmann's famous formula $S = k_B \ln(W)$ is just Shannon entropy translated to physics units. Casey said it as “entropy = force = counting at depth 0”.

Section 3.1 — Standard Second Law

$dS \geq 0$; Boltzmann $S = k_B \ln(W)$; Gibbs $S = -k_B \sum p_i \ln p_i$.

Section 3.2 — Substrate Shannon Entropy

$S = -\sum_i p_i \log p_i$ over substrate-tick $GF(128)^k$ states. Boltzmann $S = k_B \ln(W)$ recovered.

Section 3.3 — K59 Cyclotomic Chain Irreversibility

K59 RATIFIED Tuesday: 7-step cyclotomic projection on $GF(128)$ is irreversible per tick. Integrated over many ticks $\rightarrow$ macroscopic second law.

Section 3.4 — Quantum Entropy von Neumann

$S_{vN} = -\text{Tr}(\rho \ln \rho)$ substrate-cartography reading: von Neumann entropy of substrate state ρ on Bergman $H^2(D_{IV}^5)$.

Section 3.5 — Honest scope + Cross-link Vol 7

Vol 7 Information Theory provides extended Shannon entropy framework + thermodynamic-information bridging (Landauer's principle).

— Lyra, Vol 6 Ch 3 v0.3, Saturday 2026-05-23 EDT